

TMR 4190 FINITE ELEMENT METHODS IN STRUCTURAL ANALYSIS

AUTUMN 2026

EXERCISE 4

Problem 1

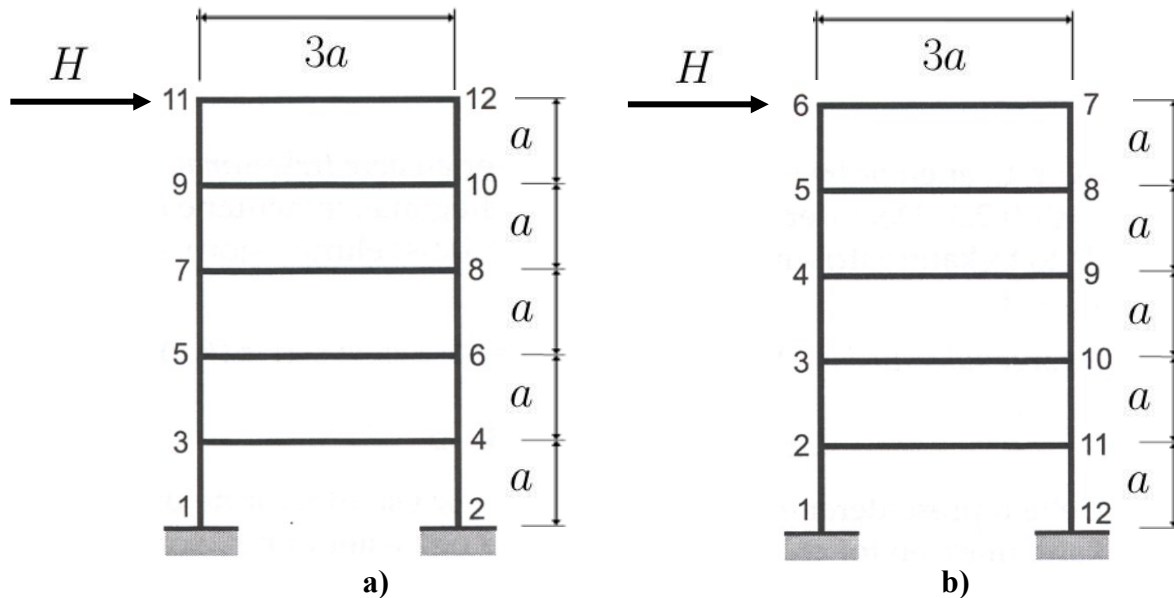


Figure 1

Figure 1 shows geometry, loading and boundary conditions of a planar frame consisting of several beam elements rigidly connected to each other and fixed to the ground. All the vertical beams have bending stiffness EI , while the horizontal ones have bending stiffness $2EI$. A horizontal force H acts on the top-left corner of the structure.

The problem parameters are given as follows:

$$a = 1 \text{ m}, EI = 10^4 \text{ Nm}^2, H = 1000 \text{ N}.$$

Use a suitable value for parameter μ , in order to neglect axial deformations.

- Using the Matlab implementation of the Matrix Method, compute the displacement vector of the structure $\mathbf{r}$ using node numbering as in Figure 1a. How much is the horizontal displacement of the node to which the force H is applied?
- Now compute the displacement vector of the structure $\mathbf{r}$ numbering the nodes as in Figure 1b. Check the horizontal displacement of the node to which the force H is applied.
- Determine the half bandwidth b_w of the stiffness matrix $\mathbf{K}$ both for cases a) and b). Use the formula:

$$b_w = n_{ndof} \times (n_{ndiff} + 1)$$

Check the calculation of b_w looking at the stiffness matrices (`>> K` in command window) and using the command `>> spy(K)`, that provides a quick overview of the sparsity of the stiffness matrix (the dots represent the non-zero elements of the matrix).

Which numbering scheme gives a more banded stiffness matrix?

- d)** Now remove all the horizontal elements but the upper one. How much is the horizontal displacement of the node to which the force H is applied?
- e)** Consider the initial geometry and the node numbering scheme a). Where would you add a single element of stiffness EI in order to minimize the horizontal displacement of node 11? How much is this displacement, in the geometry modified by you?
- f)** Consider the initial geometry and the node numbering scheme a). Add an internal hinge at node 11. How much is the horizontal displacement of node 11? Compare it with the result obtained for the original geometry (1a).

Include in your solutions the numerical results and the discussions required, the vector $\mathbf{r}$ and the plots of the deformed geometry obtained in every simulation you perform. For task c), include also the figures obtained by `>> spy(K)`.

For tasks a), b), d), e), and f), upload also the main scripts (as .m files) that you run for the simulations.

Problem 2

Derive the stiffness matrix ($\mathbf{k}_{rr}^m$) and element load vector ($\mathbf{S}_r^{\text{om}}$ for a distributed load p in positive z direction) for the 3 dof beam element in Figure 2b, by static condensation, starting from the stiffness relations of 4 dof beam element (Figure 2a).

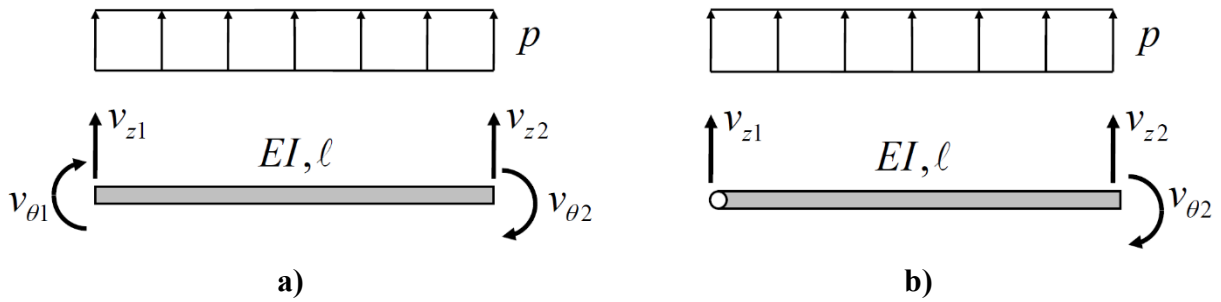


Figure 2